BUSFIN 8200: Problem Set 1

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1 Question 1

1.1 Part (a)

Assume $D_t > 0$. By definition, the gross return on equity is

$$R_{e,t+1} = \frac{P_{t+1} + D_{t+1}}{P_t}. \tag{1}$$

Taking logs,

$$\begin{aligned}
\log(R_{e,t+1}) &= \log\left(\frac{P_{t+1} + D_{t+1}}{P_t}\right) \\
&= \log\left(\frac{P_{t+1} + D_{t+1}}{P_{t+1}} \cdot P_{t+1}\right) - \log(P_t) \\
&= \log\left[\left(1 + \frac{D_{t+1}}{P_{t+1}}\right) P_{t+1}\right] - \log(P_t) \\
&= \log(P_{t+1}) + \log\left(1 + \frac{D_{t+1}}{P_{t+1}}\right) - \log(P_t).
\end{aligned}$$

Define $r_{e,t+1} \equiv \log(R_{e,t+1})$, $p_t \equiv \log(P_t)$, and $d_t \equiv \log(D_t)$. Since $\dfrac{D_{t+1}}{P_{t+1}} = e^{\log(D_{t+1}/P_{t+1})} = e^{d_{t+1}-p_{t+1}}$, we can write

$$r_{e,t+1} = p_{t+1} + \log\left(1 + e^{d_{t+1}-p_{t+1}}\right) - p_t.$$

Define $f(x) \equiv \log(1 + e^x)$, where $x = d_{t+1} - p_{t+1}$. We approximate this function around the average log dividend-price ratio $\overline{d-p}$ using a first-order Taylor approximation,

$$f(x) \approx f(\overline{d-p}) + f'(\overline{d-p})\left(x - \overline{d-p}\right),$$

so that

$$f(d_{t+1} - p_{t+1}) \approx f(\overline{d-p}) + f'(\overline{d-p})\left[(d_{t+1} - p_{t+1}) - \overline{d-p}\right].$$

Since $f(x) = \log(1 + e^x)$, we have $f'(x) = \dfrac{e^x}{1 + e^x}$, so

$$f'(\overline{d-p}) = \frac{e^{\overline{d-p}}}{1 + e^{\overline{d-p}}}.$$

Define $\kappa \equiv \dfrac{1}{1 + e^{\overline{d-p}}}$. Then $1 - \kappa = \dfrac{e^{\overline{d-p}}}{1 + e^{\overline{d-p}}}$, so $f'(\overline{d-p}) = 1 - \kappa$.

Hence,

$$\begin{aligned}
f(d_{t+1}-p_{t+1}) &\approx f(\overline{d-p}) + (1-\kappa)\Big[(d_{t+1}-p_{t+1}) - \overline{d-p}\Big] \\
&= f(\overline{d-p}) - (1-\kappa)\overline{d-p} + (1-\kappa)(d_{t+1}-p_{t+1}).
\end{aligned}$$

Define $\kappa_0 \equiv f(\overline{d-p}) - (1-\kappa)\overline{d-p}$, a constant. Since $f(\overline{d-p}) = \log(1+e^{\overline{d-p}})$ and $\kappa = \dfrac{1}{1+e^{\overline{d-p}}}$, we have $1+e^{\overline{d-p}} = \dfrac{1}{\kappa}$, so

$$f(\overline{d-p}) = \log\left(\frac{1}{\kappa}\right) = -\log\kappa, \qquad e^{\overline{d-p}} = \frac{1}{\kappa} - 1, \qquad \overline{d-p} = \log\left(\frac{1}{\kappa}-1\right).$$

Therefore,

$$\kappa_0 \equiv f(\overline{d-p}) - (1-\kappa)\overline{d-p} = -\log\kappa - (1-\kappa)\log\left(\frac{1}{\kappa}-1\right).$$

We then have $\log(1+e^{d_{t+1}-p_{t+1}}) = f(d_{t+1}-p_{t+1}) \approx \kappa_0 + (1-\kappa)(d_{t+1}-p_{t+1})$, so

$$\begin{aligned}
r_{e,t+1} &= p_{t+1} + \log\Big(1+e^{d_{t+1}-p_{t+1}}\Big) - p_t \\
&\approx p_{t+1} + \kappa_0 + (1-\kappa)(d_{t+1}-p_{t+1}) - p_t \\
&= \kappa_0 + (1-\kappa)d_{t+1} + \kappa p_{t+1} - p_t \\
&= \kappa_0 + \Big[(1-\kappa)d_{t+1} - d_t + \kappa p_{t+1} - p_t + d_t\Big] \\
&= \kappa_0 + (d_{t+1}-d_t) + \kappa(p_{t+1}-d_{t+1}) - (p_t-d_t) \\
&= \kappa_0 + \Delta d_{t+1} + \kappa(p_{t+1}-d_{t+1}) - (p_t-d_t).
\end{aligned}$$

Therefore, we have

$$r_{e,t+1} \approx \kappa_0 + \Delta d_{t+1} + \kappa(p_{t+1}-d_{t+1}) - (p_t-d_t), \qquad \kappa = \frac{1}{1+e^{\overline{d-p}}}, \qquad \kappa_0 = -\log(\kappa) - (1-\kappa)\log\left(\frac{1}{\kappa}-1\right).$$

Isolating $p_t - d_t$ and recursively substituting for $p_{t+h} - d_{t+h}$, we have

$$\begin{aligned}
(p_t - d_t) &\approx \kappa_0 - r_{e,t+1} + \Delta d_{t+1} + \kappa(p_{t+1}-d_{t+1}) \\
&= \kappa_0 - r_{e,t+1} + \Delta d_{t+1} + \kappa\Big[\kappa_0 - r_{e,t+2} + \Delta d_{t+2} + \kappa(p_{t+2}-d_{t+2})\Big] \\
&= (1+\kappa)\kappa_0 + (\Delta d_{t+1} + \kappa\Delta d_{t+2}) - (r_{e,t+1} + \kappa r_{e,t+2}) + \kappa^2(p_{t+2}-d_{t+2}) \\
&\vdots \quad \text{keep substituting } p_{t+h} - d_{t+h} \\
&= \frac{\kappa_0(1-\kappa^H)}{1-\kappa} + \sum_{h=1}^{H} \kappa^{h-1}\Delta d_{t+h} - \sum_{h=1}^{H} \kappa^{h-1} r_{e,t+h} + \kappa^H(p_{t+H}-d_{t+H}).
\end{aligned}$$

Thus, we have

$$d_t - p_t \approx \frac{\kappa_0(\kappa^H - 1)}{1-\kappa} + \sum_{h=1}^{H} \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^{H} \kappa^{h-1}\Delta d_{t+h} + \kappa^H(d_{t+H}-p_{t+H}),$$

so, letting $dp_t \equiv d_t - p_t$,

$$dp_t \approx \frac{\kappa_0(\kappa^H - 1)}{1-\kappa} + \sum_{h=1}^{H} \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^{H} \kappa^{h-1} \Delta d_{t+h} + \kappa^H dp_{t+H}. \tag{1.1}$$

Taking expectations as of time t,

$$dp_t = \frac{\kappa_0(\kappa^H - 1)}{1-\kappa} + \sum_{h=1}^{H} \kappa^{h-1} \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^{H} \kappa^{h-1} \mathbb{E}_t[\Delta d_{t+h}] + \kappa^H \mathbb{E}_t[dp_{t+H}]. \tag{1.2}$$

Suppose $H \to \infty$. Since $\kappa = \dfrac{1}{1+e^{\overline{d-p}}} \in (0,1)$, we have $\lim_{H\to\infty} \kappa^H = 0$, and by the no-bubble condition $\lim_{H\to\infty} \kappa^H \mathbb{E}_t[dp_{t+H}] = 0$. Therefore,

$$dp_t = \frac{-\kappa_0}{1-\kappa} + \sum_{h=1}^{\infty} \kappa^{h-1} \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^{\infty} \kappa^{h-1} \mathbb{E}_t[\Delta d_{t+h}]. \tag{1.3}$$

1.2 Part (b)

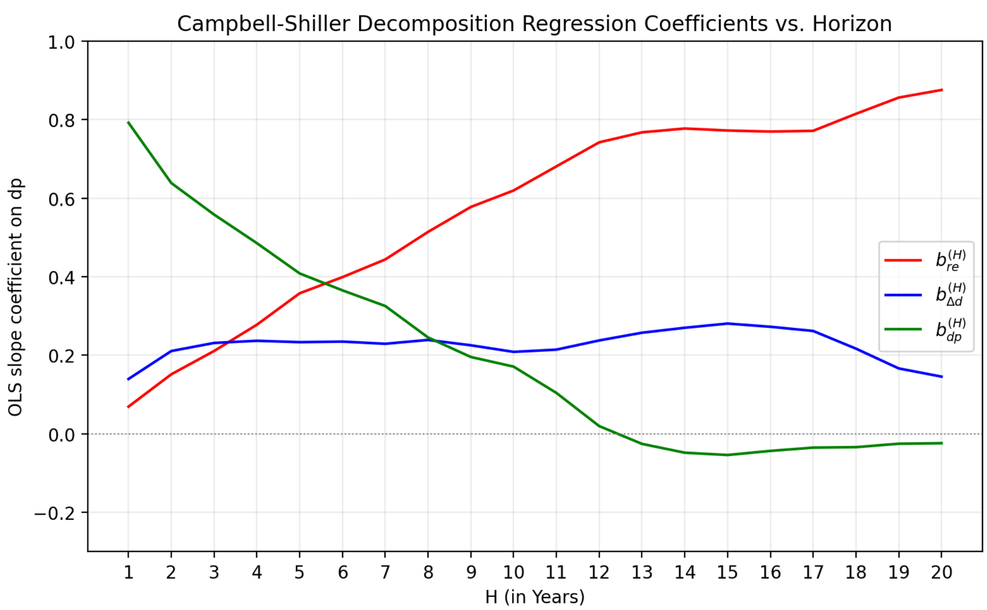

Figure 1: OLS slope coefficients of `future_return`, `change_d`, and `dp_H` on `dp`, plotted against the horizon H (in years).

From the equation

$$1 = \underbrace{\frac{\text{Cov}\left[dp_t,\, \sum_{h=1}^{H} \kappa^{h-1} r_{e,t+h}\right]}{\text{Var}[dp]}}_{b_{re}^{(H)}} + \underbrace{\frac{\text{Cov}\left[dp_t,\, -\sum_{h=1}^{H} \kappa^{h-1} \Delta d_{t+h}\right]}{\text{Var}[dp]}}_{b_{\Delta d}^{(H)}} + \underbrace{\frac{\text{Cov}\left[dp_t,\, \kappa^H dp_{t+H}\right]}{\text{Var}[dp]}}_{b_{dp}^{(H)}}.$$

This equation is trying to measure, how much each component moves when today's dp_t moves. The meaning of these three coefficients is as follows:

$$\sum_{h=1}^{H} \kappa^{h-1} \mathbb{E}_t[r_{e,t+h}] = a + b_{r_e}^{(H)} dp_t + \epsilon_t$$

The slope is $b_{r_e}^{(H)}$, and it measures the sensitivity of expected discounted future returns to today's dividend-price ratio.

Because dividend growth enters the Campbell–Shiller equation with a minus sign, define

$$-\sum_{h=1}^{H} \kappa^{h-1} \mathbb{E}_t[\Delta d_{t+h}] = a + b_{\Delta d}^{(H)} dp_t + \epsilon_t.$$

So $b_{\Delta d}^{(H)}$ captures how much variation in dp_t is associated with expected future dividend growth.

$$\kappa^H \mathbb{E}_t[dp_{t+H}] = a + b_{dp}^{(H)} dp_t + \epsilon_t.$$

$b_{dp}^{(H)}$ tells us how much today's high dp_t is simply expected to persist into the future.

$$1 = b_{r_e}^{(H)} + b_{\Delta d}^{(H)} + b_{dp}^{(H)}$$

From the graph, we can see that the contributions of expected future returns, expected future dividend growth, and the future dividend-price ratio to the variation in today's dp_t change with the horizon H. In the short run, the variation in today's dp_t is primarily driven by the future dividend-price ratio. As the horizon H increases, the contribution of expected future returns becomes more significant, while the contribution of the future dividend-price ratio declines toward zero. The contribution of negative expected future dividend growth remains relatively stable across different horizons. Also, if we add up these three coefficients, we get approximately 1, which reflects the normalized variance decomposition of today's dp_t.

1.3 Part (c)

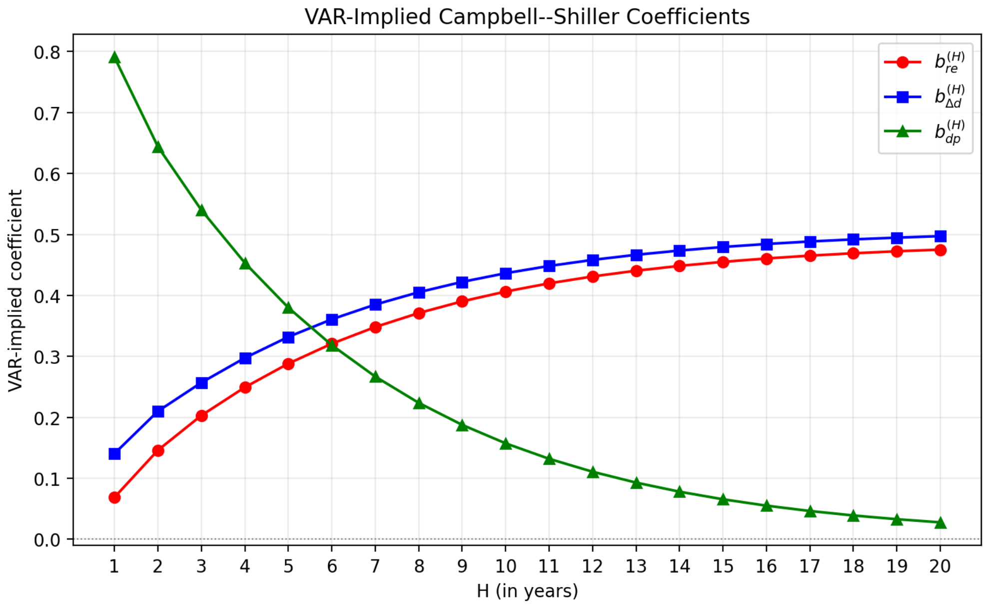

Figure 2: VAR-implied coefficients

2 Question 2

3 Question 3

4 Question 4